

Android-Powered Access Control Via Bluetooth

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Lately, there has been much talk of Android operating system, which is a touch-based interface for mobile devices like smartphones and tablets. Interfacing Android-powered devices using Bluetooth to control household appliances like fans, lights

and ACs is becoming popular amongst students and hobbyists. Learning and playing with Android-based devices is not only interesting but useful as well.

Presented here is a password-based access control system using your Android device. You can open a gate or door only when you send a valid password from your Android phone.



If you are comfortable with programming a microcontroller, you can do it easily, provided you have a Bluetooth module.

System requirements are Android-based smartphone, Bluetooth module, Freeduino board, electric strike lock YS

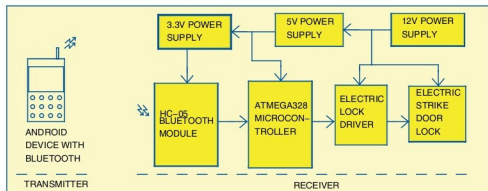


Fig. 1: Block diagram for Android-powered access control via Bluetooth

Test Points

Test point	Details
TP0	0V (GND)
TP1	12V
TP2	5V
TP3	3.3V
TP4	Low when S1 pressed
TP5	High pulses when password entered

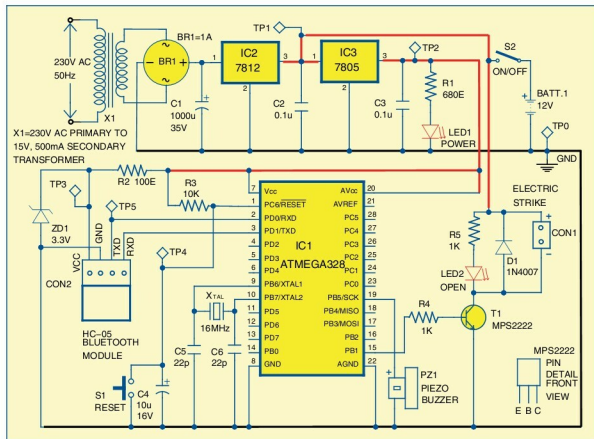


Fig. 2: Receiver circuit

130, Arduino 1.0.2 and Bluetooth SPP.

Circuit and working

Block diagram for Android-powered access control via Bluetooth is shown in Fig. 1. The system consists of a transmitter and a receiver. The transmitter is nothing but your Android smartphone or tablet with Bluetooth enabled. The receiver consists of power supplies, AVR microcontroller, Bluetooth module, solenoid driver and solenoid-based electric lock.

You can transmit control signal from your Android device to the receiver to lock